

MARBLE BAR, AustraliaWMO: **953170**Lat: **21.176S**Lon: **119.750E**Elev: **600**StdP: **14.38**Time Zone: **8.00 (E08)**Period: **01-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	50.3	53.0	7.5	8.3	83.8	13.1	10.9	79.4	20.4	67.2	18.8	69.2	3.0	140	0.450

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	25.1	111.3	67.9	109.2	68.0	107.3	68.2	78.9	86.9	78.0	86.9	77.2	87.0	8.9	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.4	146.0	80.8	76.2	139.8	81.0	75.0	134.1	81.5	43.0	86.1	42.0	86.2	41.2	86.5	95.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.0	18.9	16.6	44.7	114.9	1.7	2.6	43.5	116.8	42.6	118.3	41.6	119.8	40.4	121.7		
(5)			37.3	83.2	2.2	5.1	35.8	86.9	34.5	89.9	33.3	92.8	31.7	96.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	82.2	92.0	90.6	88.6	84.4	75.2	68.6	68.1	71.0	78.0	86.6	90.4	93.1
	DBStd	10.22	5.38	5.32	5.19	4.20	4.84	4.29	4.28	4.03	4.39	5.07	4.40	4.14
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	37	0	0	0	0	1	14	18	4	0	0	0	0
	CDD50	11744	1301	1136	1197	1033	781	559	563	650	840	1136	1212	1336
	CDD65	6306	836	716	732	583	318	123	115	189	390	671	762	871
	CDH74	90967	12824	10852	10708	7778	3252	1033	1193	2299	5166	10074	11895	13893
	CDH80	56784	8602	7067	6701	4479	1438	249	322	890	2851	6520	8051	9614
(14) Wind	WSAvg	7.3	8.2	7.7	7.2	6.5	6.9	7.1	6.5	6.7	6.7	7.8	8.4	8.5
(15) Precipitation	PrecAvg	15.3	3.1	4.4	2.3	0.8	0.9	0.8	0.4	0.3	0.0	0.1	0.4	1.8
	PrecMax	31.4	8.0	13.7	10.9	5.6	7.4	6.5	4.3	3.5	0.7	0.7	2.8	12.4
	PrecMin	6.7	0.1	0.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	6.1	2.1	3.4	2.7	1.4	1.5	1.4	0.9	0.7	0.1	0.2	0.6	2.2
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	113.6	113.2	108.9	104.3	96.2	88.6	89.1	94.0	101.2	107.7	110.6	114.3
		MCWB	68.4	68.2	69.6	67.3	62.7	63.1	59.5	60.3	62.1	65.3	65.6	67.8
	2%	DB	110.8	110.1	106.0	101.5	93.8	85.5	86.4	90.8	98.4	105.6	108.6	110.9
		MCWB	68.5	68.7	69.3	66.4	62.3	60.9	59.6	59.3	61.4	64.6	65.6	68.0
	5%	DB	108.4	106.9	103.8	99.0	90.9	83.0	83.7	88.1	96.0	103.6	106.6	108.6
		MCWB	69.6	69.8	69.2	65.9	62.1	59.5	58.7	58.2	61.0	64.3	65.6	68.1
	10%	DB	105.8	103.6	101.2	96.5	88.1	80.2	80.9	85.4	93.5	101.4	104.2	106.1
		MCWB	70.7	71.1	69.6	65.9	61.3	58.1	57.2	57.8	60.7	64.1	65.6	68.5
	0.4%	WB	79.7	79.9	79.5	76.8	72.5	73.5	69.7	72.6	81.7	72.9	76.2	78.2
		MCDB	88.7	88.3	86.1	83.3	81.1	76.2	75.1	74.0	82.8	85.9	88.8	88.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	78.6	78.4	78.1	74.9	69.3	67.5	65.7	63.7	66.3	70.3	73.6	76.6
		MCDB	87.6	87.1	85.5	84.1	80.4	75.8	76.8	80.5	83.2	88.5	88.3	88.6
	5%	WB	77.8	77.7	77.1	73.1	67.3	65.5	63.4	61.5	64.3	68.8	71.7	75.5
		MCDB	87.7	87.2	85.4	84.7	79.6	75.2	76.5	81.1	84.0	89.9	90.6	89.1
	10%	WB	77.1	76.9	76.0	71.6	64.9	63.0	60.6	59.6	62.7	67.3	70.1	74.4
		MCDB	87.7	87.3	85.7	85.2	80.2	73.8	76.0	81.0	86.6	90.4	90.1	90.2
(35) Mean Daily Temperature Range	5% DB	MDBR	22.9	22.0	21.2	23.1	23.2	22.8	25.0	28.1	29.4	27.5	26.3	25.1
		MCDBR	27.2	27.0	25.3	26.0	27.3	25.9	28.7	32.6	32.8	28.9	27.8	27.8
		MCWBR	7.4	7.3	6.5	6.8	8.8	9.3	10.9	11.9	10.5	7.9	7.4	6.9
	5% WB	MCDBR	21.1	20.7	19.2	19.8	19.1	19.0	22.8	26.5	28.4	26.4	26.0	23.5
		MCWBR	6.2	6.4	6.4	6.5	8.7	9.0	11.0	11.3	10.5	8.5	9.0	6.2
(40) Clear-Sky Solar Irradiance	taub	0.394	0.386	0.375	0.353	0.334	0.319	0.320	0.321	0.340	0.365	0.373	0.375	
	taud	2.468	2.508	2.517	2.519	2.491	2.525	2.495	2.447	2.413	2.381	2.403	2.477	
	Ebn at Noon	302	300	295	289	283	282	285	295	302	303	306	307	
	Edh at Noon	38	36	34	31	30	28	30	33	37	40	40	37	
(44) All-Sky Solar Radiation	RadAvg	2332	2226	2017	1809	1527	1383	1497	1851	2169	2407	2551	2522	
(45)	RadStd	163	176	158	87	117	107	95	57	79	84	90	122	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	+0.88	N/A	N/A	N/A	+0.86	+0.94	N/A	N/A	N/A	+284

Nomenclature: See separate page